

MONCHO UNSCHOOLING

The Maya

Unit Study

“Imagine stepping into a jungle where people read the stars, wrote in glowing picture-words, and built stone cities that still stand in Guatemala, Mexico and Belize today.”

History | Geography | Math | Language Arts | Astronomy | Biology | Art & Creativity | Movement & Body | Chemistry | Physics

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🧭 **How to use this guide:** This is not a workbook, and there's no answer key — on purpose! Every challenge is an invitation. Pick the ones that spark curiosity, skip the rest, and go find the answers by doing. **AI plans. Kids DO.** 🚀

📜 HISTORY

The Maya left their story in stone, paper and paint, and you can step right into it, one year at a time.

📜 Challenge 1: Pacal's 68-Year Path

🎯 Comparing time spans

What would your life look like if you were still in charge 68 years from now? ❓

Pacal the Great ruled the city of Palenque, in what is now Mexico, for about 68 years in the 600s AD, and was buried in the Temple of the Inscriptions. Let's walk that whole reign.

1. In a yard or on a patio, chalk a start line. Pace out 6 big steps in a straight line, one step for every 10 years. Chalk a mark at each step, and a final short mark for the last 8 years.
2. At each mark, place a small stone and chalk a label: how old Pacal could be, what might be built, and what might happen in the city.
3. Mark your own life too, on a second line. What stays the same for you and Pacal? What changes?
4. Walk the path from start to end, counting out loud.

🌱 **Make it Easier:** Pace only 4 steps, each one worth about 17 years, and label just two things at each stop.

🚀 **Level it Up:** Make a second path for the reign of a ruler from your own country's history. Compare which one lasted longer and by how many years.

🗺️ Challenge 2: Maya Time Scroll

🎯 Timelines

If the Maya story is more than 1,700 years long, where do you find it in Guatemala today?

❓

You're the museum guide, and visitors are waiting to see how the Maya story fits into time. The Maya civilization developed in Mesoamerica (southern Mexico, Guatemala, Belize, and parts of Honduras and El Salvador), and its Classic period lasted roughly from AD 250 to 900.

1. Unroll a long strip of paper (tape sheets together if needed). Mark AD 250 at the left and the current year at the right.
2. Use a ruler to space your marks: one centimeter for every 25 years works well.
3. Color the Classic period, AD 250 to 900, in a bold color.
4. Draw the years after AD 900 in a different color. Add pictures of where Maya people appear in modern Guatemala: families, languages, markets, festivals, clothing.
5. Roll the strip up like an ancient scroll and present it.

🌱 **Make it Easier:** Mark only four dates: AD 250, AD 900, the year you were born, and today.

🚀 **Level it Up:** Add three more events from Guatemala's history after AD 900. Research the dates and place them in the right spot on your scroll.

🧱 Challenge 3: Stairway of My Family

🎯 Recording history

What would your family story look like if you had to carve it into stone?❓

Maya rulers recorded history in stone, like on the Hieroglyphic Stairway, where the steps carry the story of their city. Now it's your turn to record history that lasts.

1. Pick 4 to 5 important moments from your family: a birth, a move, a celebration, a new home.
2. Draw each one as a simple picture on paper first.
3. Press a bar of soap or a block of modeling clay flat. With a toothpick or dull pencil, carve one moment per step, shaping the soap or clay into stair-like levels.
4. Place the steps in order, oldest at the bottom.
5. Brush or paint a little blue on the finished carvings.

Why: stone and baked clay don't fade or tear like paper, so carved stories can be read long after the people who made them are gone.

🌱 **Make it Easier:** Carve just 3 moments, using simple symbols instead of detailed drawings.

🚀 **Level it Up:** Add a short carved "caption" for each step using your own invented picture-symbols, then ask someone to read your stairway.

🗺️ GEOGRAPHY

Guatemala's mountains, jungles and coasts shaped where the Maya built their cities, grew their food and made their homes.

🏔️ Challenge 1: Build Guatemala from Salt Dough

🎯 Relief Mapping

Why do rivers in Guatemala flow the way they do? ❓

Millions of Maya people still live in Guatemala today, and more than 20 Mayan languages are spoken there, including K'iche', Kaqchikel, Q'eqchi' and Mam. Let's build the land they call home.

1. Mix 2 cups flour, 1 cup salt and about 1 cup water. Knead into a smooth dough.
2. Flatten it on a tray in the rough shape of Guatemala. Pile high, bumpy volcanoes and lakes on the south and west, keep the north flat and low for jungle, and leave a long smooth edge for the Pacific coast.
3. Press in three pins or toothpicks with paper flags: Guatemala City, Antigua and Flores.
4. Trickle a little water on the tallest peaks. Watch which way it runs, then draw an arrow on a paper flag showing the flow.
5. Open a free satellite view of Guatemala online with a grown-up and compare it to your model. What did you get right?

Why: Water always runs downhill because gravity pulls it toward the lowest ground, so rivers follow the valleys between the mountains.

🌱 **Make it Easier:** Skip the pins and just shape the three regions (mountains, jungle, coast), then trickle the water.

🚀 **Level it Up:** Add a river you spot on the satellite view, and test if your dough valleys send water the same way.

🌋 Challenge 2: Highland Home, Lowland Home

🎯 Comparing Regions

Would you live differently beside a lake in the mountains than deep in the jungle? ❓

Highland and lowland Maya had very different landscapes, so their daily lives were different too. You'll build both side by side.

1. On one tray, shape cool, high mountains around a dip in the middle for Lake Atitlán. Use a blue paper circle or a puddle of blue-colored dough for the lake.
2. On a second tray (or the other half of a big one), flatten the dough for the low, hot Petén jungle. Press in small green paper "trees" or crumpled tissue.
3. Place 3 tiny clay or paper houses by the lake, and 3 deeper in the jungle.
4. Write one word on a flag for each place: how people there might get water, food or travel.
5. Look at the bird on a Guatemalan coin or bill: the quetzal, the national bird, whose long green tail feathers the Maya prized. Which of your two landscapes would you expect to find it in?

🌱 **Make it Easier:** Build just the lake village with 3 houses, then talk about what the jungle might be like.

🚀 **Level it Up:** Add a trail between the two landscapes and guess how many days it might take to walk, then check your guess on a map.

🧭 Challenge 3: Plan a Trip to Tikal

🎯 Route Planning

What would you need to survive a week in the rainforest? ❓

Tikal was built in rainforest where rain falls mostly in one season, so the Maya built reservoirs and plastered surfaces to collect and store rainwater. Your family is heading there, so plan wisely.

1. Open a map and find Guatemala City and Tikal National Park near Flores.
2. Use the map scale and a piece of string to estimate the distance in kilometers between them.
3. Decide how you'd travel (car, bus or plane) and how long each way might take.
4. Make a packing list of at least 8 items for hot, wet jungle weather. Mark which ones keep you dry.
5. Pick the season you'd visit and explain why in one sentence.

Why: In a rainforest, water is plentiful in the rainy season but scarce in the dry one, which is why saving rain mattered so much to the Maya.

🌱 **Make it Easier:** Just find both places on the map, choose how to travel, and list 5 things to pack.

🚀 **Level it Up:** Add a daily budget in quetzales for food and transport, and add up the total cost of the trip.

➗ MATH

Numbers helped the Maya build cities, track the stars and run busy markets, and now you get to count like they did.

🐚 Challenge 1: The Temple Accountant's Beans

🎯 Maya numerals

How could you write every number there is with just three tiny symbols? ❓

You're the temple accountant, and the king wants his tribute counted. Use beans for dots (1), sticks or pencils for bars (5), and a small shell or coin for zero. Write each number on paper and build it with your pieces.

1. Build 3 (three beans), 7 (a stick and two beans) and 12 (two sticks and two beans).
2. Build 19, the biggest number with one set of symbols: three sticks and four beans.
3. Solve 6 + 7 by pushing the pieces together. Five beans make a new stick, so trade them in.
4. Solve 8 + 9 the same way and write both answers on paper.

Why: the Maya counted in base 20, so a number's place in the stack mattered, and the shell showed an empty place, like our zero.

🌱 **Make it Easier:** Only build numbers up to 10 and add two small numbers, like 3 + 4.

🚀 **Level it Up:** Stack a second row above your first one. The top row counts groups of 20, so one dot there is worth 20. Build 25 and 41.

📏 Challenge 2: How Many of Me Tall?

🎯 Measuring height

How many of you would have to stand on each other's heads to reach the top of a giant jungle temple? ❓

Tikal, in the Petén region of northern Guatemala, was a major Maya city. Its Temple IV rises about 70 m, one of the tallest pre-Columbian structures in the Americas.

1. Cut a strip of paper or string the same length as your height. Ask a helper to measure it with a tape or ruler (about 130 cm is common at your age).
2. Divide 7,000 cm (70 m) by your height to find how many of you would stack up Temple IV.
3. Divide 3,000 cm (30 m) by your height to compare with a 30 m pyramid like El Castillo.
4. Outside, lay your strip end to end and count out the stack. Mark the finish line with chalk, then walk beside it and see how far you travel.

🌱 **Make it Easier:** Use a round height of 100 cm. Then 70 m is exactly 70 of you, and 30 m is 30 of you.

🚀 **Level it Up:** Find the difference between the two stacks. Then measure a tree or building near you and add it to the line.

⚙️ Challenge 3: Two Wheels, One Great Cycle

🎯 Cycles and multiples

If two calendars start together today, how long before they land on the same pair of dates again? ❓

The Maya used a 260-day sacred calendar (Tzolk'in) and a 365-day solar calendar (Haab'). Together they form a cycle of about 52 years called the Calendar Round. Build two paper wheels to test it.

1. Draw a small wheel with 13 sections and a large wheel with 20 sections, like the two wheels inside the 260-day Tzolk'in. Mark them with numbers and day names of your own.
2. Pin both wheels together at the center with a brad or a pencil.
3. Turn both one notch at a time for 20 days. Are both wheels back at their starting spots together? Multiply 13 × 20 to find when they will be.
4. Compare: 260 days × 73 turns = 365 days × 52 turns = 18,980 days. That is how long until the same date pair meets again.

Why: 260 and 365 share only the factor 5, so the two cycles take a long time to line up.

🌱 **Make it Easier:** Use two small wheels of 3 and 4 sections and count how many turns until both are back at the start.

🚀 **Level it Up:** Make a third wheel with 7 sections for the days of our week. How many days until all three line up?

📜 LANGUAGE ARTS

Long before screens and keyboards, Maya writers filled bark-paper books and stone walls with glowing picture-words, and stories from the Maya world are still told in Guatemala today.

📜 Challenge 1: The Royal Scribe's Secret Letter

🎯 Writing Systems

Could you write a whole message without using a single letter of the alphabet? ❓

You are the royal scribe, and the king needs a secret message that only his trusted reader can decode. The Maya wrote with glyphs, a system mixing picture-signs and sound-signs, and scribes copied them into folding bark-paper books called codices. Only 4 Maya codices are known to survive (Dresden, Madrid, Paris, Grolier).

1. Invent 6 picture-signs for things, such as sun, jaguar, corn or river.
2. Invent 4 sound-signs for syllables, such as "ma," "ya," "ka" and "to."
3. Write a 2-sentence message using only your signs.
4. Make a decoder key on a separate paper, then hand the letter to a family member and see if they can crack it.

🌱 **Make it Easier:** Use only picture-signs for 5 simple words, with no sound-signs at all.

🚀 **Level it Up:** Make a longer message that combines signs, like the Maya did, with a picture-sign and a sound-sign inside one block.

🎭 Challenge 2: The Hero Twins Come Alive

🎯 Oral Storytelling

How do you tell a story so well that people can see it without a book? ❓

The Popol Vuh is a K'iche' Maya book of creation stories, including the tale of the Hero Twins and the making of people from maize. The best-known written copy was recorded in the 1700s by Father Francisco Ximénez in Chichicastenango, Guatemala. Now it is your turn to tell it.

1. Find a children's version of the Hero Twins story at the library, or ask a grown-up to read one aloud.
2. Make 3 or 4 stick puppets from paper, craft sticks or spoons.
3. Split the tale into a beginning, a middle and an end, and practice your own words for each part.
4. Use different voices for each character, and tell the story aloud to your family.

🌱 **Make it Easier:** Tell only the part of the story you liked best, using just one puppet.

🚀 **Level it Up:** Add a narrator voice, sound effects and a surprise ending that still fits the story's meaning.

🗣️ Challenge 3: Word Detectives of Guatemala

🎯 Word Origins

How many Maya words are hiding in the things you say every day? ❓

Many Guatemalan place names come from Maya languages, like Xela, short for Xelajú, the K'iche' name for the city of Quetzaltenango.

1. Interview 3 family members or neighbors, and ask: "Which Maya words or place names do you know, and what do they mean?"
2. Write each word, who told you, and its meaning.
3. Look up any mystery words in a dictionary, a book or with a family member who speaks a Maya language.
4. Sort your collection into groups: animals, foods, places and names.

🌱 **Make it Easier:** Collect just 5 words from one family member, and draw a tiny picture for each.

🚀 **Level it Up:** Find a word that exists in both a Maya language and Spanish, and write a short guess about how it traveled between them.

🔭 ASTRONOMY

The Maya watched the sky so closely that they could predict where Venus would appear years ahead of time, and you can start sky-watching too.

🔭 Challenge 1: Track Venus

🎯 Sky observation

Why does the brightest "star" in the sky sometimes vanish for weeks and then return?❓

You are a Maya sky-watcher keeping a record of Venus. The Maya tracked it carefully, and the Dresden Codex holds tables that predict its appearances over cycles of about 584 days.

1. At dusk or just before dawn, find a spot with a clear view of the horizon, or open a free sky app such as Stellarium or Sky Guide and search for Venus.
2. Draw a big circle on paper and divide it into 8 equal slices, like a pizza. The whole circle stands for 584 days.
3. Mark today's date on the rim. Write "seen" or "not seen" and sketch where Venus sits in the sky, high or low and east or west.
4. Check again every few days for two weeks and add each observation to the wheel.

Why: Venus circles the Sun inside Earth's orbit, so from Earth it sometimes shines in the evening, sometimes in the morning, and sometimes hides in the Sun's glare.

🌱 **Make it Easier:** Make only 4 slices and check just three times this week. Write "yes" or "no" for whether you saw Venus.

🚀 **Level it Up:** Count how many days Venus is hidden before it comes back. Then predict the date of its next appearance and check whether you were right.

🔭 Challenge 2: Build the Shadow Snake

🎯 Light and shadow

How can the Sun draw a snake on the side of a stone pyramid? ❓

At Chichén Itzá in Mexico, the pyramid El Castillo is about 30 m tall. At the spring and autumn equinoxes, the Sun makes a shadow that looks like a snake sliding down its stairs. Build a small version and find the angle that makes it.

1. Cut strips of cardboard and stack 3 or 4 boxes, each smaller than the one below, to make a stepped pyramid.
2. Darken the room and place the pyramid on a table.
3. Hold a lamp or phone flashlight about 40 cm away and shine it from the side, low and high.
4. Look for the angle where the steps cast a zigzag, sawtooth shadow. Try at least 5 different angles.
5. Write down or sketch which angle worked best, and note how far you held the lamp.

Why: Each step blocks light from the step below, so a low angle of light makes a row of triangles that looks like the scales of a snake.

🌱 **Make it Easier:** Use a simple 3-step pyramid and a flashlight. Just try low, middle and high, and circle the one that looks most like a snake.

🚀 **Level it Up:** Try a pyramid with more steps, then fewer. Does the snake shadow change? Measure the angle of your lamp with a protractor and write it down.

🔭 Challenge 3: Sun Marker

🎯 Measuring shadows

If the Sun seems to move all day, why does a shadow at noon still change from week to week? ❓

1. Find a flat, sunny spot in your yard. Push a straight stick upright into the ground, or fix it in a pot of soil, and make sure it doesn't lean.
2. Use a pebble or chalk to mark the tip of the shadow at noon, or when the shadow is shortest.
3. Measure the shadow from the base of the stick to the mark with a ruler or string.
4. Do this at the same time for 5 days or more, and write each day's date and length in a table.
5. Look at your table. Is the shadow getting longer, shorter or staying the same?

Why: Earth is tilted, so the Sun's path across the sky climbs higher or lower as the seasons change, and shadows shrink or stretch with it.

🌱 **Make it Easier:** Measure only three days and compare with the words longer or shorter instead of numbers.

🚀 **Level it Up:** Also measure the shadow at 9 a.m. and 3 p.m. on one day. Draw all three shadows from the same stick and describe how they point.

🌿 BIOLOGY

Plants, bugs and birds are the living world the Maya depended on every single day, and you can study them right where you are.

🌽 Challenge 1: Grow a Milpa in Pots

🎯 Plant growth

Why would farmers plant three different crops together in one patch instead of separate rows? ❓

You're the milpa farmer, and your field is a few pots. Maize was the central crop of the Maya, grown with beans and squash in milpa fields, and the Popol Vuh says the first people were shaped from maize.

1. Fill a big pot or bucket with soil and plant 3 corn seeds in the middle.
2. When the corn is about 15 cm tall, plant 2 bean seeds near the stalks.
3. Plant 1 squash seed near the edge of the pot.
4. Put the pot outside in a sunny spot and water it every day or two.
5. Once a week, draw what you see: how tall is the corn, and where is the bean growing?

Why: the bean vine can't hold itself up, so it climbs the sturdy corn stalk, while the wide squash leaves shade the soil and keep it from drying out.

🌱 **Make it Easier:** Plant only corn and one bean in a cup, and draw it just two times.

🚀 **Level it Up:** Measure each plant with a ruler every week and make a line graph of corn height versus bean height.

🔍 Challenge 2: Square-Meter Safari

🎯 Food chains

How many different living things can squeeze into a patch of ground no bigger than a bath towel? ❓

Find a backyard, park or even a single patch of grass.

1. Mark one square meter with string or four sticks.
2. Crouch down and look for every living thing: ants, beetles, weeds, moss, spiders. Use a magnifying glass, or zoom in with a phone camera.
3. Sketch five of your finds in a notebook.
4. Sort your sketches into three piles: eats plants, eats other bugs, eats dead stuff.
5. Draw arrows from what gets eaten to what eats it.
6. Count the creatures you found and write the number down.

Why: each arrow shows energy moving from one living thing to the next, and the dead-stuff eaters turn leftovers back into soil for plants.

🌱 **Make it Easier:** Find and sketch just three living things, and sort them into only two piles.

🚀 **Level it Up:** Visit the same square meter at a different time of day and compare who shows up.

🪶 Challenge 3: Quetzal Feather Detective

🎯 Adaptation

What makes a bird's body just right for living high up in a misty cloud forest? ❓

The quetzal lives in the cloud forests of Guatemala, and you're going to investigate it like a scientist.

1. Look up photos of a quetzal in a book or online. Notice the shimmering green feathers, the long tail, the crest, the round head and the strong feet.
2. Go outside and find a feather from a pigeon, chicken or any other bird. If you can't find one, use a photo of a feather.
3. Look at both feathers closely, with a magnifying glass if you have one. Compare the color, the shape and the fuzzy parts.
4. Gently run your finger along the feather and watch the tiny barbs zip back together.
5. Write down two ways the quetzal's body helps it live among misty, leafy trees.

Why: the tiny barbs hook together like a zipper, making a smooth surface that sheds water and helps a bird fly.

🌱 **Make it Easier:** Compare only the color and size of the two feathers, and draw them side by side.

🚀 **Level it Up:** Look for a third kind of feather, like a fluffy down feather, and explain which part of the bird's body it probably comes from.

🎨 ART & CREATIVITY

The Maya turned cloth, walls and stone into pictures, and you can make your own in the same spirit.

🧵 Challenge 1: Weave Your Village Motif

🎯 Weaving patterns

How can a few simple crossing threads become a picture that tells people who you are? ❓

Around Lake Atitlán in Guatemala's highlands, weavers in villages such as Santiago Atitlán and San Juan La Laguna still use backstrap looms and natural dyes. You're the village weaver, and your cloth needs its own motif.

1. Cut a piece of cardboard about the size of your hand. Make 9 small notches along the top and bottom edges.
2. Wind string up and down through the notches. These are your warp threads.
3. Sketch a motif on scrap paper, like a bird, a zigzag or a flower, using only straight lines and blocks.
4. Cut strips of colored paper or yarn. Weave them over and under the warp, changing colors to build your motif row by row.
5. Push each row up snugly against the one before it before starting the next.

🌱 **Make it Easier:** Weave with wide paper strips through a sheet of paper cut into slits, and just make a stripe pattern.

🚀 **Level it Up:** Add a second motif that repeats across the width, and try a border of tiny diamonds along the edges.

🖌️ Challenge 2: Paint a Festival Mural

🎯 Mural painting

What would a whole village look like on festival day if you could only see it painted on one long wall? ❓

Think about a big celebration, with dancers, music, food and crowds. Today you paint it in earth colors, the way painters have done for thousands of years.

1. Tape paper pieces together, or use a long roll, until your sheet is at least three times longer than it is tall.
2. Make your paints from what you can find. Mash berries or beet slices for reds, steep tea or coffee for browns, rub green leaves and spinach for greens, and mix dirt with a little water for tan.
3. Sketch your scene lightly in pencil, from left to right, like a story: people arriving, dancing, playing music, sharing food.
4. Fill in big shapes first, then add small details like feathers, flowers and patterns on clothing.
5. Leave some space between figures so each one stands out.

Why: Plant and earth colors come from natural pigments, which are tiny colored particles that stay in the paint after the water dries.

🌱 **Make it Easier:** Paint just one dancer and one drum on a single sheet of paper.

🚀 **Level it Up:** Add a border of repeating patterns around your mural and show a sunrise on one end and a night sky on the other.

🗿 Challenge 3: Carve a Ruler in Profile

🎯 Relief sculpture

What makes a face look powerful when you can only see it from the side? ❓

Maya artists often showed leaders in profile, which means from the side, with the nose, forehead and chin forming a bold outline.

1. Roll a ball of clay or take a bar of soap, and flatten it into a slab about as thick as your thumb. (If you use soap, just use the flat side of the bar.)
2. Sketch a side-view face with a pencil or a toothpick: a strong nose, a sloping forehead, a firm chin.
3. Press and carve out the shape so the face rises up from the background. Use a spoon handle or toothpick to scrape away the extra.
4. Build a tall headdress above the head by adding rolls and shapes of clay, such as feathers, flowers or shells. (If you use soap, carve the headdress shapes into the soap instead.)
5. Add details with the toothpick: lines on the headdress, an earring, a necklace.
6. Let it dry, or leave soap as it is, and display it on a shelf.

🌱 **Make it Easier:** Draw the profile first on paper, cut it out, and press it into the clay as a guide.

🚀 **Level it Up:** Add a second figure facing the first one and carve a border of tiny patterns around both.

🏃 MOVEMENT & BODY

Move like the Maya: play their famous ball game, climb like a pyramid explorer, and dance like a festival jaguar.

🏐 Challenge 1: Hips-Only Ball Game

🎯 Teamwork

How hard is it to keep a ball alive when your hands are not allowed to help?❓

The Maya played a ball game called pitz with a solid rubber ball, and stone ball courts still stand at Copán in Honduras and Iximché in Guatemala. Let's play a safe version!

1. Mark a court with tape, chalk or string, about 5 big steps wide and 10 long, with a center line.
2. Make two teams. Use a soft ball, like a beach ball or a rolled-up sock ball.
3. Hit the ball only with hips and thighs, no hands. Each team tries to send it over the center line so the other team can't return it before it lands.
4. A team scores a point when the ball lands on the other side. Play to 7, and mark every point on a scoresheet.

🌱 **Make it Easier:** Allow the ball to bounce once before the next hit, and let players use their knees too.

🚀 **Level it Up:** Add a rule that the ball must be touched by 3 different players before it crosses the line.

🪜 Challenge 2: Pyramid Stair Climb

🎯 Counting Systems

Could you count your way to the top of a pyramid using the Maya number system instead of ours? ❓

Maya numbers use dots for 1 and bars for 5, so 7 is one bar and two dots.

1. Find a staircase at home, a park or a school, and imagine it is a pyramid with a temple at the top.
2. Climb up 10 steps, then walk down carefully. After each round, write the round number in Maya dots and bars on a card.
3. Do 5 rounds, going faster each time if it feels good.
4. Add up your total steps, such as 5 rounds × 10 steps = 50, and write that number in Maya numerals.

Why: Climbing stairs makes your leg muscles and your heart work harder, so your heart beats faster to send them more oxygen.

🌱 **Make it Easier:** Climb only 5 steps per round, and just draw dots and bars for the round numbers 1 to 3.

🚀 **Level it Up:** Climb 20 steps per round and write each running total, like 20, 40, 60, in Maya numerals.

🐆 Challenge 3: Jaguar and Monkey Festival Dance

🎯 Rhythm and Coordination

How can your whole body tell a story without using a single word? ❓

You're the festival dancer, and the village is waiting for tonight's animal dance!

1. Make a rhythm with stomps and claps, such as stomp, stomp, clap, clap. Repeat it until it feels steady.
2. Invent a jaguar move: slow, low and sneaky, with powerful steps.
3. Invent a spider monkey move: swinging arms, quick hops and a long tail wiggle.
4. Join your moves with the stomp-and-clap rhythm into a dance of about 8 counts for each animal.
5. Practice with a family member or friend, then perform the full dance together.

🌱 **Make it Easier:** Pick only one animal and repeat its move 4 times with the same rhythm.

🚀 **Level it Up:** Add a third animal, like a macaw or a turtle, and have your group dance in a circle, switching animals on a signal.

🧪 CHEMISTRY

Maya builders were practical chemists, cooking limestone, grinding pigments and roasting cacao beans.

🧱 Challenge 1: Which Wall Coating Survives the Rain?

🎯 Fair testing

Which mixture will protect a wall best when water hits it? ❓

1. Cut three equal cardboard squares to be your "temple walls." Label them A, B and C.
2. Mix three coatings: A) flour and water paste, B) plaster of Paris (an adult helps, and do not breathe the powder), C) salt dough made from 2 parts flour, 1 part salt and a little water.
3. Spread each coating on its wall in a layer about as thick as a coin. Let them dry overnight.
4. Prop all three walls up at the same angle. Spray each one 10 times from the same distance, using a spray bottle.
5. Look at each wall. Which one stayed smooth? Which turned soft, cracked or washed away?
6. Rank the three coatings from best to worst.

Why: Maya builders made lime plaster by heating limestone, then used it to coat buildings and make smooth surfaces for painting murals. A good coating has to resist water, which is why testing matters.

🌱 **Make it Easier:** Test only two coatings, flour paste and salt dough, and spray each wall 5 times.

🚀 **Level it Up:** Add a fourth wall coated with white school glue mixed with a little flour, and count how many sprays it takes before each coating starts to break down.

🎨 Challenge 2: The Pigment Maker's Fade Test

🎯 Observing change

Which natural color will still be bright after washing and sunshine? ❓

1. You are the pigment maker for a new city, and your colors must last. Gather red cabbage, turmeric powder and a cooked beet.
2. With an adult, chop the cabbage and beet, and place each in its own bowl with a little hot water. Mix the turmeric with a few spoonfuls of warm water in a third bowl. Wait 10 minutes, then strain.
3. Dip a strip of white paper towel or cotton cloth into each dye. Let all three dry on a plate.
4. Cut each strip in half. Rinse one half of each color under the tap while rubbing it gently, then leave it to dry.
5. Tape the other halves to a sunny window for 3 days.
6. Compare every washed and sunned strip with the original color. Which stayed brightest? Which faded the most?

Maya blue was famous because it stayed bright for centuries, and your test shows why that was so special.

🌱 **Make it Easier:** Test just two dyes, beet and turmeric, and only do the washing test.

🚀 **Level it Up:** Add a squeeze of lemon juice to one half of each dye, and see if acid changes the color before the fade test begins.

🍫 Challenge 3: Whisk Up a Frothy Maya-Style Chocolate

🎯 Mixing and observing

What makes a hot chocolate drink grow a foamy top? ❓

Cacao beans were valuable to the Maya, used to make a bitter, frothy chocolate drink (often with chili or maize) and sometimes used as currency. Make your own cooked, kitchen-safe version and compare it to theirs.

1. With an adult, warm 1 cup of milk in a pot on the stove until it is steaming, not boiling.
2. Pour it into a heatproof jug. Stir in 1 tablespoon of unsweetened cocoa powder and a pinch of cinnamon. For a Maya-style kick, add one tiny pinch of mild chili powder.
3. Whisk fast for 1 minute, or roll a wooden spoon between your palms in the jug. Watch the foam appear.
4. Taste it. Then stir in a little honey and taste again.
5. Write your tasting notes: How bitter was it? How thick was the foam? Which version would a Maya trader like?

Why: Whisking pushes tiny air bubbles into the milk, and the proteins and fats in milk wrap around them so the foam holds.

🌱 **Make it Easier:** Skip the chili, warm the milk in the microwave with an adult, and just whisk and describe the foam.

🚀 **Level it Up:** Make two cups, one whisked for 20 seconds and one for 90 seconds, and time how long each foam lasts before it disappears.

⚛️ PHYSICS

The Maya shaped rubber, stone and rain into tools and cities, and you can test the same forces at your kitchen table.

🏀 Challenge 1: Bounce Battle

🎯 Measuring Rebound

Why would a whole civilization care so much about a ball that bounces well? ❓

1. Tape a ruler or measuring tape straight up a wall, with zero at the floor.
2. Pick three balls: a rubber one, a tennis ball and a ball of clay. Rolled-up socks work if you have no tennis ball.
3. Drop each ball from 100 cm, with your eyes level with the rebound, and film it on a phone if you can. Replay it slowly to read the height.
4. Drop each ball three times and write down how high it bounced each time.
5. Find the average rebound for each ball and put them in order, best bouncer first.
6. The Maya made their ball from latex tapped from rubber trees, sometimes mixed with morning glory vine juice to make it bouncier. Which of your balls acts most like theirs?

Why: A bouncy ball squishes when it hits the floor and stores energy, then springs back and gives most of it back. Clay stays squished, so its energy turns into heat instead of motion.

🌱 **Make it Easier:** Test only two balls and drop each just once. Say "higher" or "lower" instead of measuring.

🚀 **Level it Up:** Drop each ball from 50 cm and from 100 cm. Does a rebound at double the height come back exactly double?

🧱 Challenge 2: Stack an Arch

🎯 Balance and Support

How far can a layer of stones stick out over empty space before gravity wins? ❓

1. Collect 20 or more small blocks, flat stones or thick books, and clear a flat table.
2. Build two matching towers about 15 cm apart, with a gap between them.
3. Add the next layer so each block sticks out a little farther toward the middle than the one below it. Maya builders stacked stones this way, which is called a corbel arch.
4. After each layer, measure the overhang with a ruler. Keep going until the stack wobbles, then write down the biggest overhang that stayed up.
5. Try to close the gap at the top with one flat block. Can your stack do it?
6. Now try a different way: lean two blocks together over the gap so they hold each other up. Compare how stable this is to your stepped stack.

Why: A block stays up while its center of mass is still over the block beneath it. Stick it out too far and its weight pulls it down.

🌱 **Make it Easier:** Use only 8 blocks and a smaller gap of about 8 cm.

🚀 **Level it Up:** Add a second tower of weights on the back ends of your overhanging blocks. How much farther can you stick out now?

🌧️ Challenge 3: Catch the Rain

🎯 Slope and Flow

Would a steeper slope catch more rain or send it rushing right past your basin? ❓

1. Press a ring of modeling clay into the bottom end of a baking tray to make a small basin, leaving a gap where water can enter.
2. Rest the other end of the tray on a stack of books to make a ramp. Start with a 2 cm stack.
3. Pour 200 ml of water slowly over the top, as if it were rain, and then measure how much collected in the basin.
4. Make the ramp steeper with a 5 cm stack, then a 10 cm stack. Repeat the same pour each time and record the amounts.
5. Make a bar chart of your results and find the slope that caught the most.

Why: On a steep slope, water gains speed and can splash, overshoot, or miss the basin. On a gentle slope it moves slowly and stays on target.

🌱 **Make it Easier:** Test only two slopes, one low and one high, and say which basin holds more water.

🚀 **Level it Up:** Cover the ramp with foil, then cloth. Does a rougher surface change how much water the basin catches?